

(2 Hours)

[Total Marks: 60]

- N. B.: (1) **All** questions are **compulsory**.
 (2) Make **suitable assumptions** wherever necessary and **state the assumptions** made.
 (3) Answers to the **same question** must be **written together**.
 (4) Numbers to the **right** indicate **marks**.
 (5) Draw **neat labeled diagrams** wherever **necessary**.
 (6) Use of **Non-programmable** calculators is **allowed**.

1. Attempt **any two** of the following: 12

- Explain six types of foundational languages used for Artificial Intelligence briefly.
- Describe five state-of-the-art AI technologies today.
- Describe different types of environments applicable to AI agents.
- Define a Learning-based agent and explain its four conceptual components with a diagram.

2. Attempt **any two** of the following: 12

- Describe the fundamental rules and constraints used in 8-Queens problem formulation and mention its components.
- Formulate 8-puzzle problem for the following scenario and also mention the steps and path cost required to reach goal state.

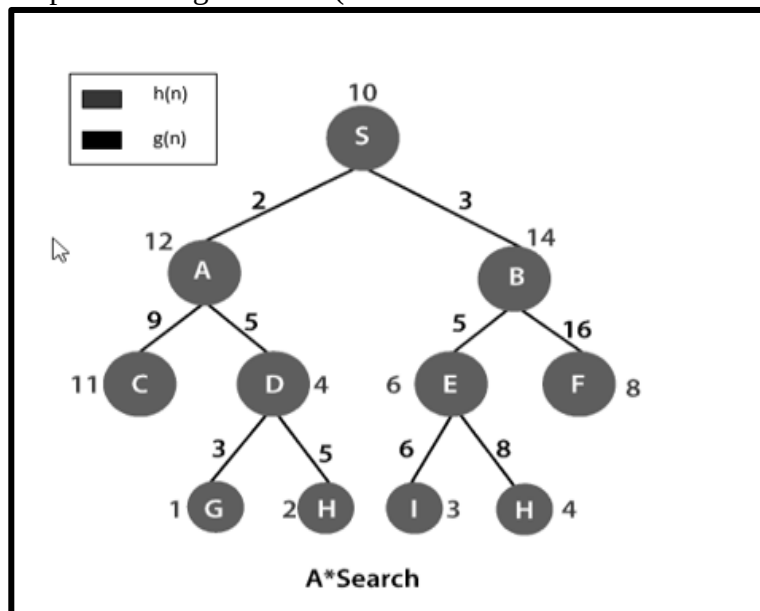
1	2	3
4	8	-----
7	6	5

Initial State

1	2	3
4	5	6
7	8	-----

Goal State

- Explain the A* algorithm, its properties, and evaluate a given search tree using A* to find the shortest path to the goal node. (**S is the root node and G is the goal node**)

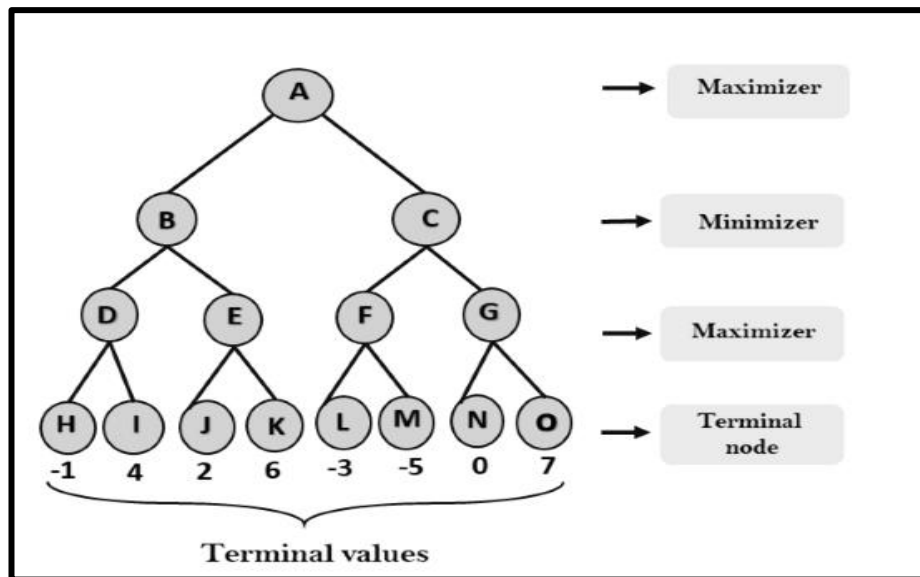


- Identify and summarize the limitations and solutions of the hill climbing search algorithm.

3. Attempt any two of the following:

12

- Define adversarial search, explain techniques for optimal solutions, and describe elements of Game Playing search.
- Describe the min-max algorithm and evaluate a given game tree using min-max to find the value of a node.



- Describe horn and definite clauses.
- Describe five connectives of Propositional Logic with an example.

4. Attempt any two of the following:

12

- Define Quantifiers, explain their different types with syntax and examples.
- Define unification and explain the process of unification.
- Compare and contrast Forward and Backward chaining reasoning methods.
- Define Resolution, summarize the steps for Resolution and convert FOL into Conjunctive Normal Form (CNF) using provided statements.
 - Convert to FOL.
 - Convert FOL to CNF.
 - Prove that "Is Someone Smiling?" Using Resolution.
 - Draw Resolution tree.

Statements:-

- All people who are graduating are happy.
- All happy people smile.
- Someone is Graduating.

5. Attempt any two of the following:

12

- Explain two different types of algorithms used for planning as state space searches.
- Explain multi-agent planning with different types of strategies.
- Differentiate between STRIPS and ADL Planning languages.
- Write a short note on "The Internet Shopping World."